

shinyplanr — Smarter Collaborative Spatial Planning

The Challenge

Countries are racing to meet global biodiversity commitments — including protecting 30% of land and sea by 2030 under the [Kunming-Montreal Global Biodiversity Framework](#). But planning where to act is complex, slow, and often gridlocked. Stakeholder processes stall. Technical tools are inaccessible. Expert-led workshops are expensive and time-consuming. Meanwhile, ecosystems cannot wait. There is a fundamental disconnect between the spatial planning tools that scientists use and the decision-makers who urgently need them.

What is *shinyplanr*?

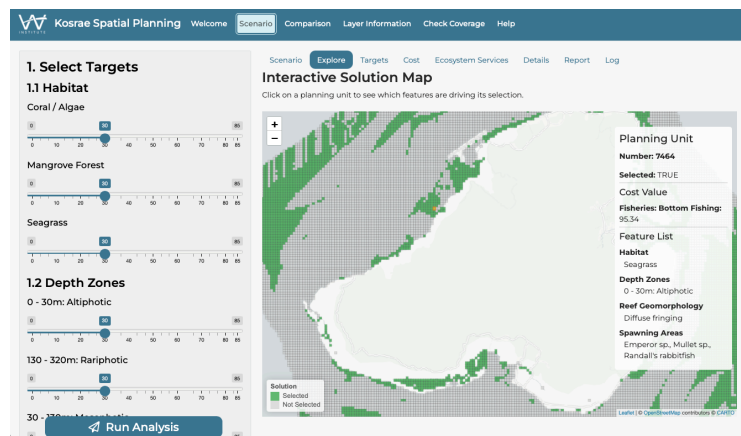
shinyplanr is a free, open-source web application that puts world-class spatial planning directly into the hands of governments, NGOs, and communities - no coding, no specialist training required.

Built by researchers at The University of Queensland and powered by the [prioritizr](#) package, *shinyplanr* delivers rigorous, transparent spatial plans in real time. Users set biodiversity targets, choose cost layers, and add constraints - and the app finds the most efficient protected area network that meets their goals, in seconds.



Crucially, *shinyplanr* is generic and transferable: a single codebase can be customised and deployed for any country, region, or jurisdiction - from the coral reefs of Fiji to the deep waters of the Weddell Sea.

What Makes *shinyplanr* Different?



Real-time decisions. Most scenarios run in seconds. Stakeholders explore, adjust, and iterate on the spot without waiting for expert analysis.

Transparent and user-driven. Users control targets, costs, and constraints themselves. More learning, more trust, faster buy-in from all parties.

Fewer workshops, faster consensus. Stakeholders explore plans independently, reducing costly and time-consuming expert-led meetings.

Climate-smart by design. Planning can be steered towards climate refugia - areas that will retain biodiversity longer under future warming - making conservation investments resilient to climate change.

Compare trade-offs side by side. Two scenarios can be run simultaneously, with maps, charts, and summary statistics compared in parallel, so the consequences of different choices are immediately visible.

Downloadable evidence. Maps, spatial files, target-achievement charts, and comprehensive HTML reports can all be downloaded instantly for reports, policies, and stakeholder presentations.

Scalable across any geography. Deployed in small island states such as Kosrae (Federated States of Micronesia) and vast regions such as the Weddell Sea, *shinyplanr* adapts to any scale, data and question.

Proven in the Field

shinyplanr is already making a real difference through partnerships with the [Waitt Institute](#), the [Norwegian Polar Institute](#), [Earth Sciences New Zealand](#) and the [University of Exeter](#). Recent and planned deployments include:

- Weddell Sea
- Kosrae, FSM
- Yap, FSM
- Fiji
- Kiribati
- Côte d'Ivoire
- Vanuatu

Who Is It For?

- **Governments** seeking transparent, repeatable, and defensible planning processes
- **NGOs** facilitating stakeholder workshops or advising on protected area design
- **Research teams** communicating spatial planning outputs to non-technical audiences
- **Community and indigenous organisations** participating in decisions about their environment

A Broader Vision: Science Meets Stakeholders

shinyplanr sits at the heart of a broader philosophy: cutting-edge science is only as powerful as its ability to reach the people who need it. Our team has a proven track record of translating complex, data-heavy research into interactive, accessible tools - and **shinyplanr** is a core-element of that approach.

We have also built [MiCO \(Migratory Connectivity in the Ocean\)](#), a Shiny app that provides access to synthesised global tracking data for over 100 migratory marine species - from sea turtles and seabirds to whales and tuna. MiCO bridges the gap between tracking data and the international frameworks that depend on them.

We have also developed [The Climate Explorer](#), an interactive tool that lets planners visualise future ocean conditions and understand how climate change will reshape marine ecosystems. This feeds directly into climate-smart **shinyplanr** deployments, where decisions are grounded in future projections, not just present-day data.

Together, these tools reflect a deliberate strategy: use world-class science to build open, transferable, web-based platforms that empower stakeholders to make better decisions for nature.

Who we are

We are a team of university researchers advancing spatial planning for real-world conservation. Our mission is to support biodiversity outcomes through transparent, research-driven tools tailored to local contexts. We co-design planning solutions, with our partners, that support biodiversity conservation, restoration, and sustainable ocean use under a changing climate. We have expertise in developing rapid, effective and inclusive spatial plans and are continually improving our tools based on user feedback and emerging science.



**Assoc. Professor
Daniel Dunn**

A marine conservation scientist with 20+ years advising on area-based management for international bodies.



**Professor Anthony
Richardson**

A mathematical ecologist with 25+ years of research experience in climate change ecology and conservation science.



Dr Jason Everett

A spatial ecologist and data scientist developing tools for designing protected areas and analysing climate change impacts.